Geminio

Language-Guided Gradient Inversion Attacks in Federated Learning

Phase 1 & 2 Progress Report

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Jiao et al., ICCV 2025

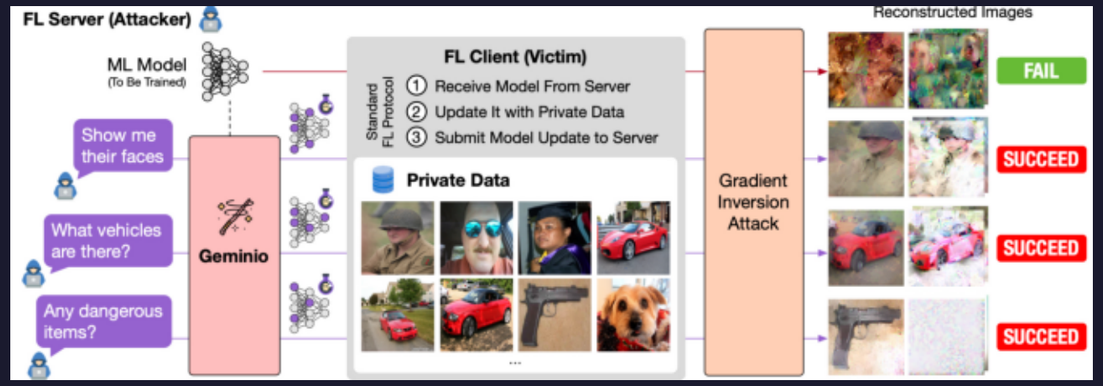
Background & Attack Mechanism

Federated Learning Threat Model:

- Clients train on private data, share only gradients
- Malicious server inverts gradients to reconstruct images
- Problem: Existing GIAs fail at large batch sizes (>=64)

Geminio's Key Contribution:

- Uses CLIP (VLM) to enable natural language queries targeting specific private data
- Reshapes the model's loss surface so query-matching images produce dominant gradients
- Enables targeted reconstruction from 128-sample batches



Loss Surface Reshaping:

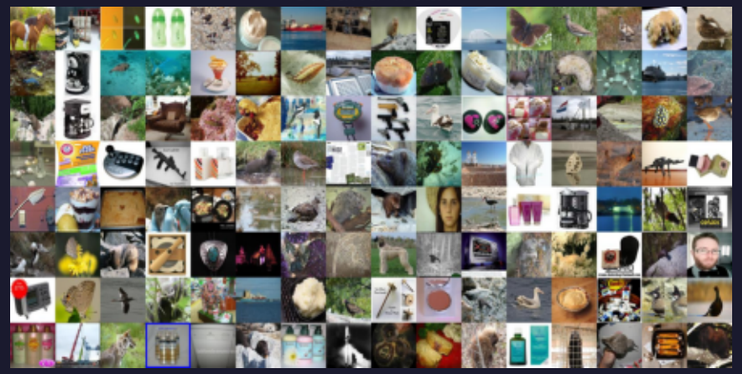
L = E[loss_i * (1 - clip_similarity_i)]

Query-matching images -> amplified gradients

Non-matching images -> suppressed gradients

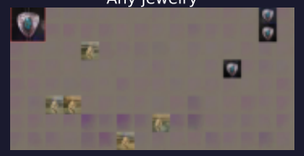
Experimental Results — ImageNet (128 Private Samples)

Ground Truth

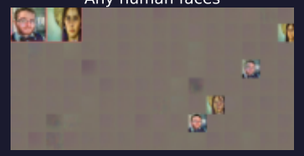


Baseline GIA

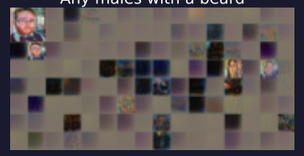
"Any jewelry"



"Any human faces"

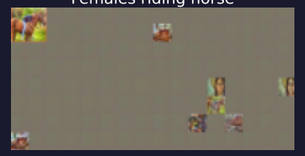


"Any males with a beard"



"Any guns"

"Females riding horse"



Reconstruction Loss Comparison

Method	Loss	Imprv.
Baseline GIA	0.1828	–
"Any jewelry"	0.0965	47.2%
"Any human faces"	**0.0750**	**59.0%**
"Males w/ beard"	0.1097	40.0%
"Any guns"	0.0984	46.2%
"Female on horse"	0.1222	33.2%

Key Findings:

- Baseline: all 128 images unrecognizable
- Geminio: query-matched images clearly reconstructed; others suppressed (gray)
- Best: "human faces" (59% lower loss)

Next Steps — Healthcare Application

Completed (Phases 1–2):

- ✓ Paper review — understood attack mechanism, VLM guidance, loss reshaping
- ✓ Full pipeline reproduced on ImageNet (VLM embeddings → training → attacks)
- ✓ Debugged & documented issues (dependency conflicts, missing configs)

Phase 3 — Healthcare Dataset Application:

- Apply Geminio attack to medical images (MRI, X-ray, patient records)
- Test if patient data can be exposed despite FL privacy guarantees
- Replace CLIP with medical VLM (BiomedCLIP) for better alignment
- Example queries: "Any chest X-ray showing pneumonia"

Future Directions:

- Propose defense mechanisms (not covered in original paper)
- Explore applicability to DoD / mission-critical settings
- Differential privacy, gradient clipping, secure aggregation